The Design of Device Monitoring System Base on Big Data Algorithm of Quality Management System

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Abstract

This paper introduces how equipment can be properly managed by means of device monitoring system base on quality management system. To sufficiently control equipment reserve and supply, protect and maintenance, calibration and verification, it separates and combines quality management system concept to complete whole chain tracking management of equipment business. This article shows that information technology and quality management system able to improve device monitoring efficiency while apply theory to practice.

CCS Concepts

• **Information systems** → Information systems applications; Computing platforms.

Keywords

Quality management, Equipment management, Equipment maintenance

ACM Reference Format:
Shou Yuan, Chen Chen, Jianyu Li, and Jingyu Zhen. 2024. The Design of Device Monitoring System Base on Big Data Algorithm of Quality Management System. In *2024 5th International Conference on Computer Science and Management Technology (ICCSMT 2024), October 18–20, 2024, Xiamen, China.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3708036.3708070

Quality Management System as one of the most credible system to enhance execution ability and management level, it participated in the process of modern industrialization to a great extent[1], and help managers optimize business processes meanwhile increase work efficiency from the perspective of standardization and customization.

While safety of people's lives and property with a highly frequency reported in governors' speech, government department invest amount of resource to develop meteorological institution, and as a foundation of weather forecast, meteorological observation equipment management and status become an important research indicators.

There are more than 17,000 automatic meteorological observation stations[2], 97 national meteorological observation stations, 14 new generation radar, and countless observation machines cover temperature, humidity, air pressure, rainfall, atmospheric composition and so on[7]. Considering calibration and verification demand and risk of equipment damage, meanwhile meteorological equipment department requires 1 year inspection, an efficient system that able to provide equipment fault information, calibration information, and warehouse storage information may greatly help user reduce management cost and time.

However, due to large numbers of devices, it is impossible to monitor devices manually. In this paper, it studies and develops a system that can automatically detect meteorological observation equipment accurately according to equipment status information by read network transmission electric signal. This design refers to the structure of provincial meteorological observation network, and integrates equipment storage system and discusses advantages and disadvantages of current equipment management mode.

1 Key points and logic design

1.1 Big Data Algorithm

Big data as the most useful information technology developed these years, taken as a foundation supporting function of an information retrieval system. Its algorithm to what extent optimizes information retrieval results, distinguish information importance, and improve information system structure. Considering a big data information system requires a stable information flow structure that prevent

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ICCSMT 2024, October 18–20, 2024, Xiamen, China
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ACM ISBN 979-8-4007-0999-9/24/10
https://doi.org/10.1145/3708036.3708070

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Figure 1: Raw Data of Meteorological Observation.

Figure 2: Quality Management System Definition.

deadlock database stack, its I/O requirement, information review, and data analysis efficiency become the most important issue to consider.

The device monitoring system owns three characteristics, batch data processing, streaming data processing system, interactive data processing. Batch data processing store massive amounts of data, use massive data to develop appropriate patterns, and ultimately generate effective countermeasures to achieve business goals. The information system of device monitoring receive and storage device sensor information every per second, like Figure 1 raw information, determine working status of different types of sensors based on threshold settings of each different device. According to basic I/O requirement and threshold algorithm, all sensor collected information will be received in central server to finish first filtering process. In order to distinguish useful information, central server adopts binary search algorithm to set valid observation total interval, its primary algorithm works like Table 1. By repeatedly dividing observation data half of portion until narrow most accurate consequences, binary search algorithm provides a possible method to compare and calculate objective values of meteorological factors in a region, that improve government decision-making accuracy regarding various consequences brought about by weather factors. After binary search algorithm finish data filtering, I/O requirement begins to intervene and adopt extreme value filtering to judge collected data and store the data that meets meteorological observation standards in central provincial server, and store doubtful data in manual review area.

To filter data accurately, EM-Training (Expectation-maximization) algorithm and Binary Search are firstly introduced to categorize and filter. Given that every sensor has its own unique metrological calibration procedures, EM-Training guarantee that data collected by each sensor judged and collected data within calibration range. Afterwards central server double check collected data then store in database.

1.2 Quality Management Process

Quality management system refers to the management system of command and control organization in terms of quality. The quality management system is a system decision-making model established within organization, and it is a strategic decision to achieve quality goals with systematic and necessary quality mode [3]. In national meteorological administration, national, provincial, city, and country meteorological bureau constitute a central vertical management mode, which has a great advantage that all of bureau able to reflect national order rapidly, however during implementation process its operation procedures or results different from expected assignments. Thus it is important to set a standard that all of staff able to refer to.

As shown in Figure 2, Quality management system is managed by principles, and principles are divided from issues resolution [4], integration, environment, and so on. During all of principles, a certain and clarify quality manual book is the priority to help user identify importance and working process. The quality manual book refines working process in accordance with different institutions into operation guidance books that can provide specific guidance steps and references. To help user easily understand the conception of quality management system and behavior as quality management system requires, internal auditor training is support taken as the priority to promote. Furthermore, inspection, management review and business improvement are important process of quality management system as shown in Figure 3.

Table 1: Measure Range of Meteorological Observation Element.

Measure Range	Temperature	Air Pressure	Rainfall	Humidity
Numerical Value	±0.2°	±0.3hPa	±0.3mm	±4%RH

Figure 3: Quality Management System Process.

To illustrates the business precisely and accurately, quality manual book, which contains all of supporting documentation required for a business system and details how business process work, compiled by experts from all aspects of business process and will be revised and improved on an basis according to actual needs, ensure that all of information in manual book is up to date and realistic to work [5]. Quality manual book usually contains three layers to establish a total and instruction structure, quality manual, management document, and work instruction. Quality manual focus on description of the cause of overall goal, and explain the cause of relevant important components. Management document accomplish with logic of work process, highlight working authority and logic of responsible party to enable quality management user easily identify working process and responsibility manager. Work instruction majority content is specific steps and supporting documents for the job. Unlike other documents it is written from the perspective of managers, work instruction provides technical staff a reference book with operation information and procedures, which allow technical staff can complete work even though first time doing similar work.

Internal audit is an important part of quality management system. In order to check whether operation of quality system is effective, organization train a group of technical personnel with professional and technical literacy to study and participate in assessment to obtain a third-party certified internal auditor certificate. A internal audit organized by top leader and authorized highest level of rights to review and post issues [5]. To guarantee professionalism, effectiveness and standardization, internal audit manual will be prepared according to the business situation and management structure of each organization. By establish a professional internal audit team, a organization would able to self-correction and possess an ability to continuously improve their own business [6].

Management review as an important work after internal audit, it aims to critical thinking and review what have done from organization and what can be fixed or improved.

1.3 Information Management

Information is a activity in which human to plan, organize, lead and control information resources in order to effectively develop and utilize information resource by means of modern information technology. In equipment monitoring and control system, there are many information management theory and technology used to gain most useful information.

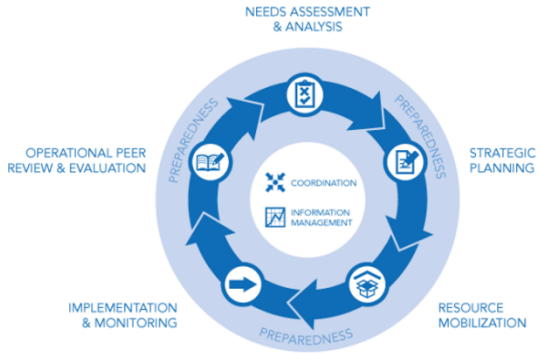

Figure 4: Information Management Life cycle.

As shown in Figure 4, to highlight most useful information and save user time, theme tree and data base is combined to provide most concise user interface but most detailed data background. Theme tree is to organize information resources layer by category according to a certain pre-determined conceptual structure [4,7]. User selects layer by layer by browsing and traverses layer by layer until they find information they need, and directly find corresponding network information resources through information clues. Its advantage is simple and easy to use, information retrieval based on a certain category classification, and good scalability and strict systematization, However, its disadvantage is it have to establish a category system in advance, and cannot design a complicated structure in case user have to invest much time to train. In order to minimize impact of disadvantage of theme tree, search engine is taken in equipment management system to help user identify issues [8].

Data base is a warehouse that organize, stores and manages data according to its data structure[9], it is a mainstream and indispensable information mode in current information system. Normal information system owns function of storage, interception, security, backup, etc. With development of information technology and demand of massive information processing, online data base technology adopt dual backup operation of physical and virtual server, distributed technology is increasing adopted in modern database technology to speed up data inquiry and avoid unnecessary error.

For an equipment information monitoring system operating under quality management system, all of information transmission, input and output process have to follow standardized principle by means of information key points audit. The information system can screen and display relevant information of equipment by filling in fault repair sheet.

2 Design of Device Monitoring and Control

In order to ensure accuracy and effectiveness of weather forecast, meteorological department established various types of devices to

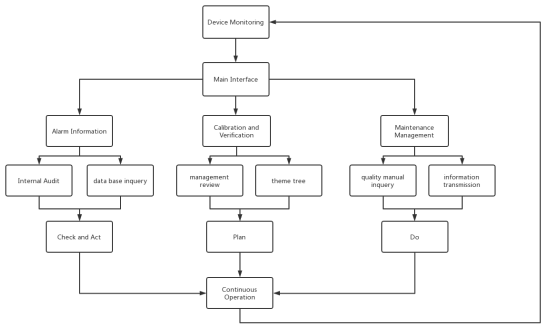

Figure 5: Logic Diagram.

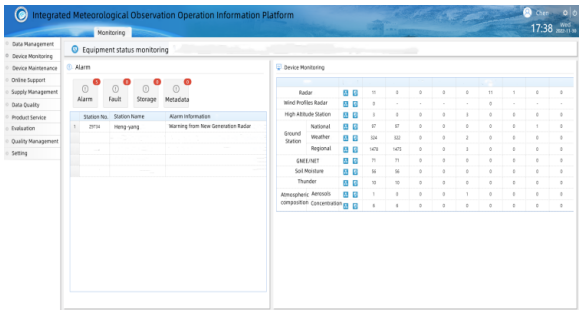

Figure 6: Equipment Monitoring Interface.

obtain meteorological observational data, including radar, high altitude observation data, satellite remote sensing observation data, etc. Thus how to apply observation data to weather forecast effectively, and how to complete forecast and inversion of past weather conditions becomes an important subject of meteorological equipment department.

As the most important function adopted in equipment management, device monitoring and control able to help user identify fault equipment and directly arrange replacement of spare parts and manpower arrangement. Its logic have to designed simple and clear, like Figure 5. In order to reduce response time, device monitor system is divided into three levels, provincial, municipal, and county-level, which allow user from different administrative area find fault accurately. To avoid unnecessary association cost, provincial, municipal, and county-level account number has been given different permissions that only can monitor equipment at this level.

Figure 6 shows two parts of monitoring interface, alarm information and equipment working information. Alarm information interface adopts a blank panel, shows alarm information one by one only when central server cannot accept observation data or other fault cases, it may help user locate faulty equipment, then sent remote repairing command or restart equipment. Considering information system have to follow accuracy, timeliness and effectiveness principle, alarm information associate with social media text able to response fault information and notify equipment maintenance staff simultaneously. The equipment working information

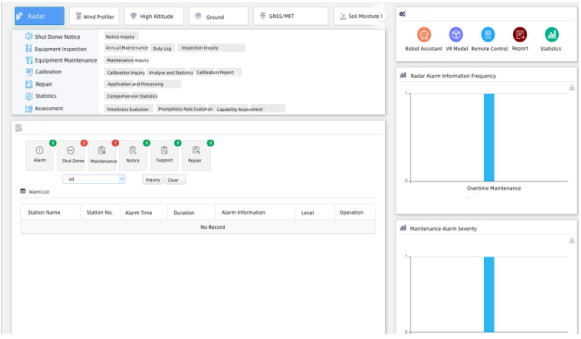

Figure 7: Maintenance Sheet.

demonstrates all of equipment information base on its observation element and type. To provide most useful information, equipment working information shown in a table with accurate information, such as regional, time, and information of maintenance. Besides, equipment working status can be checked or displayed by means of map, which make it easier for provincial user finish statistics of fault cases. And monitoring interface designs four link to storage, fault, and alarm interface.

The equipment maintenance interface divided into three parts, like in Figure 7, maintenance information and statistic requests, and technical support programs. With big data and many cases save in central service, it is convenient for equipment management system summarize fault information and make a con-vincible prediction on equipment working status. Maintenance information is highlighted in the main screen and classified alarm information and daily maintenance information according to equipment type. By quality management request, there are seven message notification in these section designed to satisfy whole process tracking, shutdown notice, equipment maintenance support, equipment maintenance, calibration and verification, invalid parts management, query, and comprehensive evaluation were ensured to guarantee quality management system established, implemented and maintained. In this part every section has its sub-section that convenient user finish equipment work, such as shutdown notice query, annual maintenance query, maintenance report query and so on. Statistic reports covers all kinds of fault alarm information and demonstrates from various equipment. Fault alarm information in this section provides detail information that allow user aware of equipment series number, contact information of maintenance personnel, maintenance order generation and tracking, etc. Which gives country-level management department a simple and clearly troubleshooting process to complete equipment maintenance procedures as soon as possible. Technical support programs focus on education and case study. According to quality management request 7.2 capacity, the organization should identify the capabilities of personnel under their control whose work affects performance and effectiveness of the quality management system. Technical support programs are constitute of maintenance robot, VR simulation model display, remote instrument display, and export report. Maintenance robot is constituted of typical cases and logic design of equipment, which provides user a clearly insight of problem solving. VR(Virtual Reality) simulation is made of Unity and it establish a virtual

environment simulates a national observatory. Besides, it simulates a few types of fault cases include sensors, electric circuit, signal transmission, etc. Remote instrument display enables user input operation command to finish easy operation such as reset, information feedback, and so on. Whereas information of equipment fault noted in this section, relevant link will be shown when user try to check typical fault cases according to key words or specific sensors.

By introduce big data technology and information management in equipment monitoring, user may gain a lot of information from statistics. From calibration and verification, it is able to find out whether detected sensor facing the situation of unable to work properly by reading gap between detector data and standard data. By introducing maintenance information, support staff able to calculate which equipment or sensor have highest frequency failures and causes of failures. By introducing monitoring data, the statistics of fault repair orders can help users find type of fault, time of occurrence, and duration of fault, etc. Through data collected by different functional modules, the system can evaluate and predict equipment with frequent failures, and so as to sort out equipment failure factors and complete research and judgement of functional requirements for redevelopment.

3 Conclusion

In conclusion, it is proved that information technology can be used in quality management system and enhance effectiveness of work. After investigate 25 users of device monitor and control system, feedback demonstrates that user have better understanding and save a lot of time compared from information system introduced. However, manager users also declare that they may have to invest much more time to finish internal audit and cost more human resource when reach specific job deadline.

There are advantages when designs a information system introducing quality management system. However, there are many risks such as information system encounter massive information visits, insufficient server bandwidth, etc. Progress in information models must be accompanied and considered by the progress of information technology, which is an essential direction of future scientific and technological progress.

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